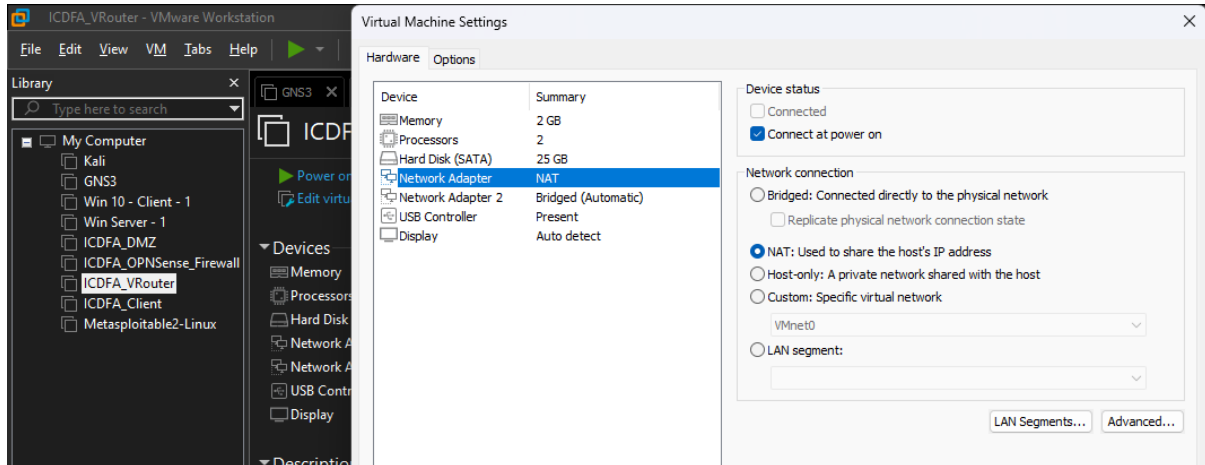


FULL NAME	MUSA SHUAIBU
REGISTRATION NUMBER	C11/26/FCDF/17153
UBUNTU	ICDFA provided
VM HOSTNAME	Icdfa-nslab-client-v1

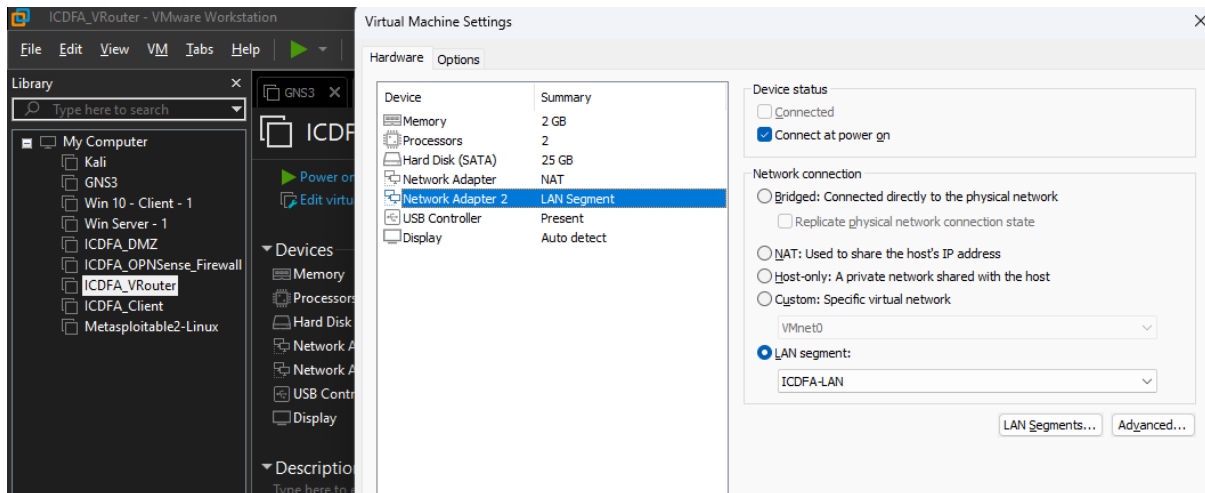
PART A: PREPARING VM WORKSTATION:

1.0 VRouter WAN interface



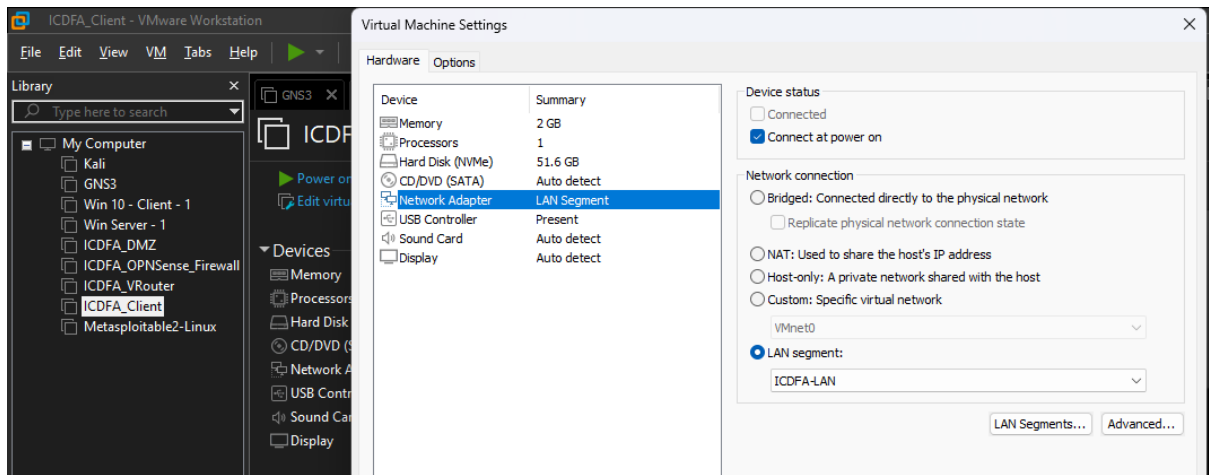
Brief Explanation: the Nat interface allow the VRouter to be able to access the internet through the host

1.1 VRouter LAN interface



Brief Explanation: The LAN segment isolate the interface to private network and it cannot be use to access the internet

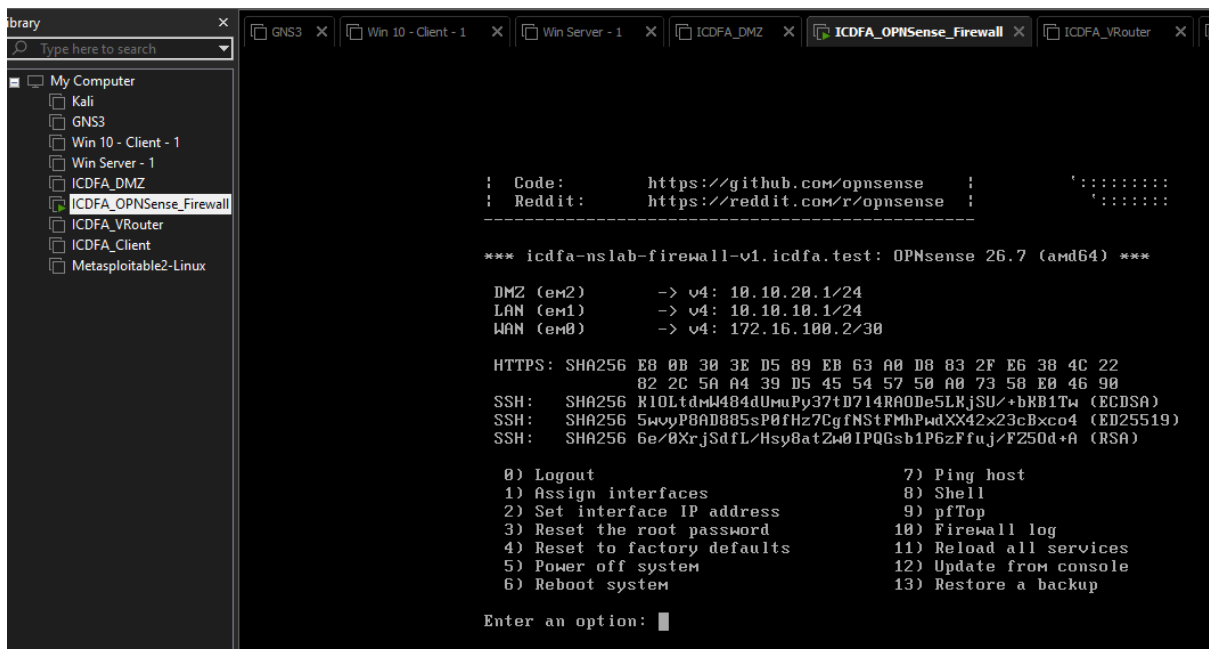
1.2 Client VM



Brief Explanation: The LAN segment isolate the interface to private network and it cannot be use to access the internet

PART B: VERIFYING OPNsense INTERFACE

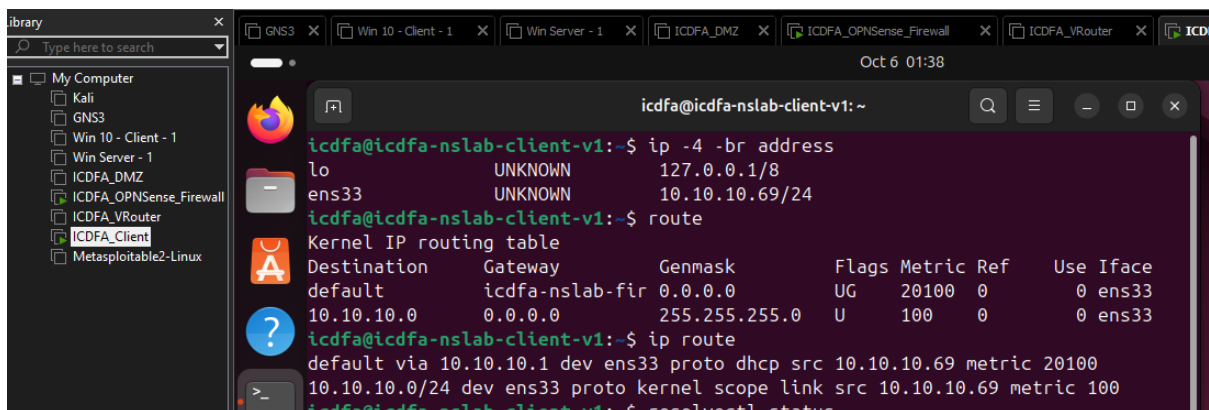
2.0 OPNsense Firewall



Brief Explanation: The LAN interface has an ip addr of 10.10.10.1 /24 which is a private class A ip address

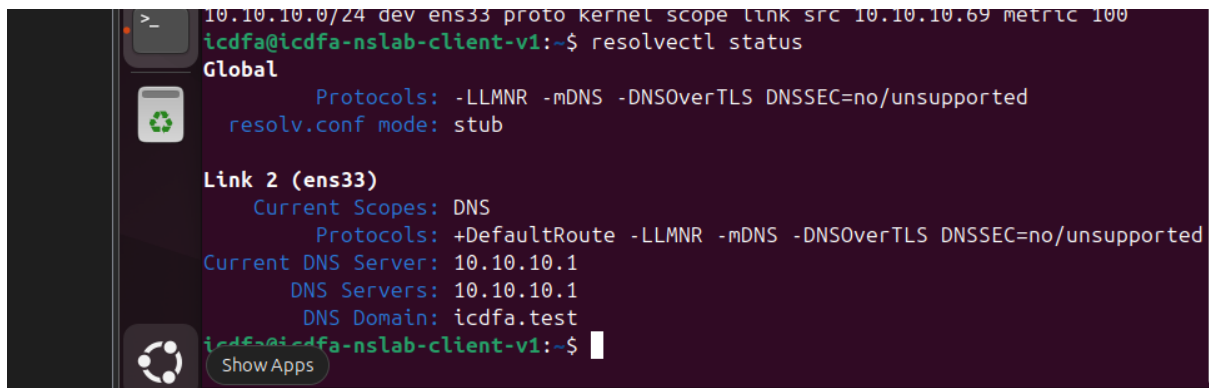
PART C: VERIFYING UBUNTU ADDRESS

3.0 Client VM



Brief Explanation: The Ubuntu client has obtained IP address (10.10.10.69) from the firewall DHCP server which has a default gateway of 10.10.10.1 and the command `ip -4 -br` displays its brief information for v4 only.

3.1 Client VM



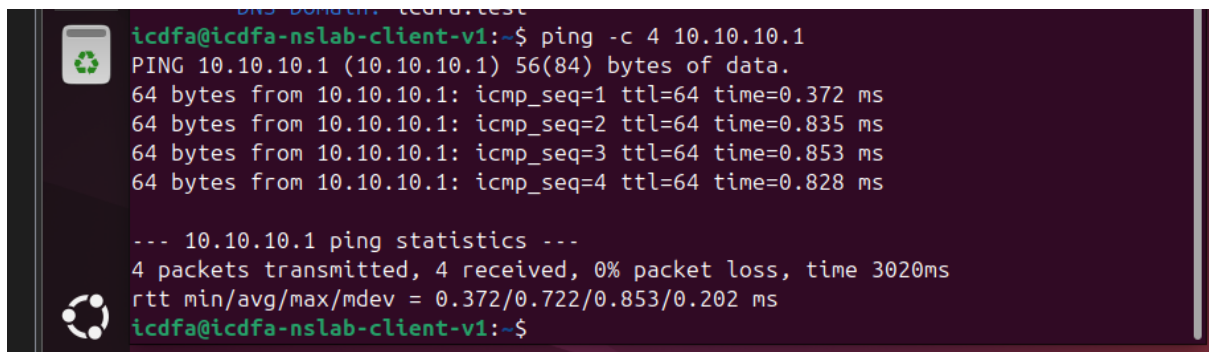
```
10.10.10.0/24 dev ens33 proto kernel scope link src 10.10.10.69 metric 100
icdfa@icdfa-ns-lab-client-v1:~$ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 2 (ens33)
    Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    Current DNS Server: 10.10.10.1
    DNS Servers: 10.10.10.1
    DNS Domain: icdfa.test
icdfa@icdfa-ns-lab-client-v1:~$
```

Brief Explanation: The `resolvectl status` is used to check or ask who translates names for me.

PART D PERFORMING CONNECTIVITY TEST

4.0 Testing connectivity

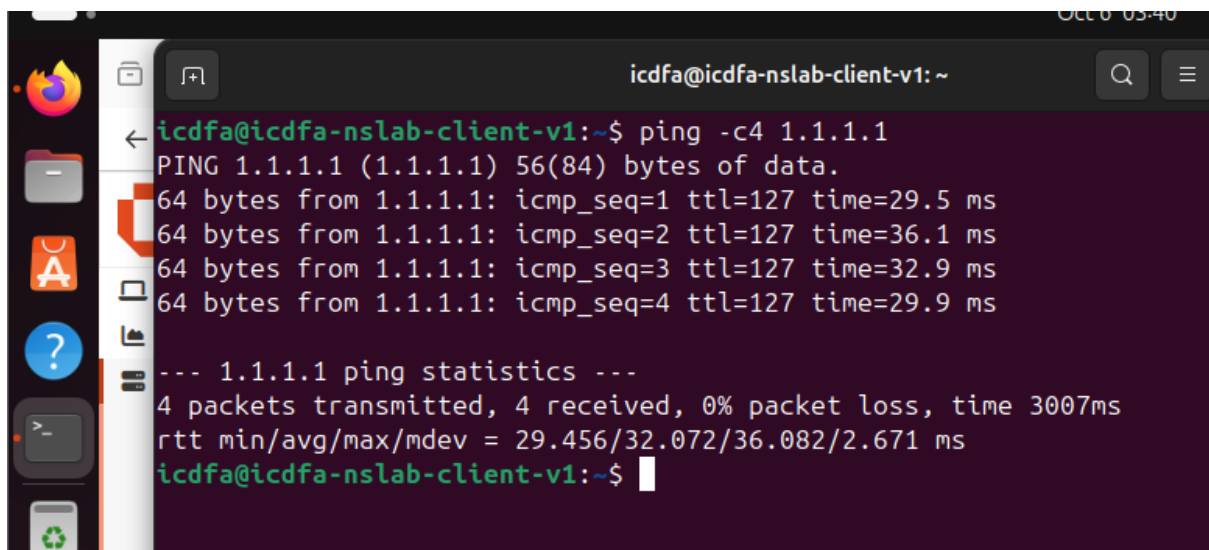


```
icdfa@icdfa-ns-lab-client-v1:~$ ping -c 4 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data:
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=0.372 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=0.835 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=0.853 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=64 time=0.828 ms

--- 10.10.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3020ms
rtt min/avg/max/mdev = 0.372/0.722/0.853/0.202 ms
icdfa@icdfa-ns-lab-client-v1:~$
```

Brief Explanation: This shows that the Ubuntu can reach the firewall LAN interface which is its gateway.

4.1 Testing connectivity



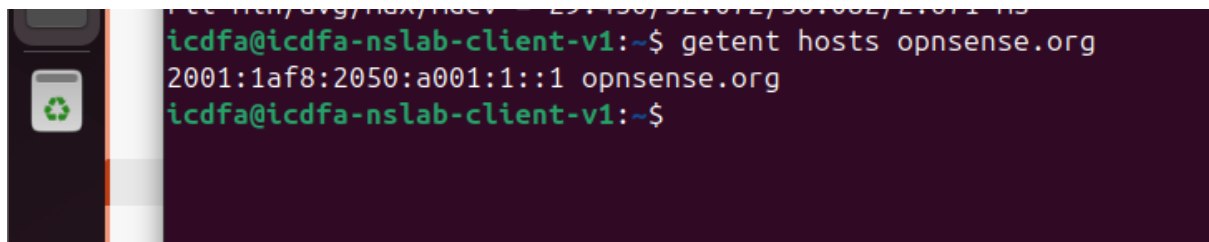
```
icdfa@icdfa-ns-lab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data:
64 bytes from 1.1.1.1: icmp_seq=1 ttl=127 time=29.5 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=127 time=36.1 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=127 time=32.9 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=127 time=29.9 ms

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 29.456/32.072/36.082/2.671 ms
icdfa@icdfa-ns-lab-client-v1:~$
```

Brief Explanation: This shows that the client VM can communicate with the internet but before that I have to troubleshoot it, the WAN interface has been assigned a static sub IP of 172.16.1.0 and default gateway of 172.16.1.1 while my VMnet8 has a sub IP of 192.168.163.0.

with gateway 192.168.163.2 therefore I change the WAN interface to obtain ip from dhcp and I log into the GUI of the firewall and disable the previous WAN interface config, so that the new obtain IP from DHCP will be the upstream gateway

4.2 Testing connectivity



```
icdfa@icdfa-ns-lab-client-v1:~$ getent hosts opnsense.org
2001:1af8:2050:a001:1::1 opnsense.org
icdfa@icdfa-ns-lab-client-v1:~$
```

Brief Explanation: This means Ubuntu successfully translated the hostname opnsense.org into an IP address. Which is ip v6 address

4.3 Testing connectivity

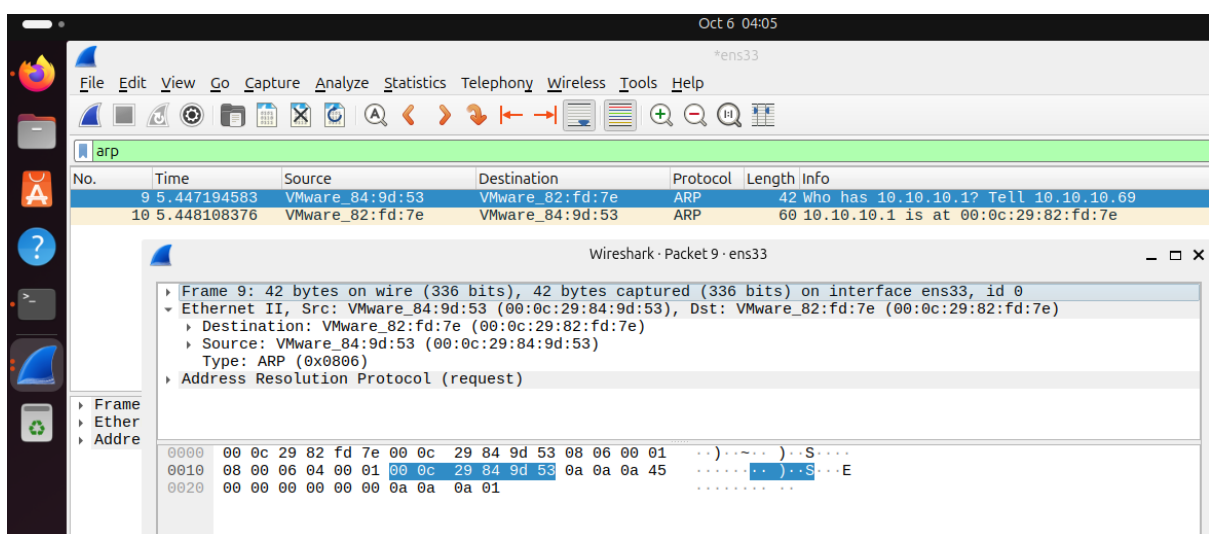


```
icdfa@icdfa-ns-lab-client-v1:~$ curl -I https://opnsense.org
HTTP/1.1 200 OK
Date: Tue, 06 Oct 2026 01:52:18 GMT
Server: Apache
Link: <https://opnsense.org/wp-json/>; rel="https://api.w.org/", <https://opnsense.org/wp-json/wp/v2/pages/199>; rel="alternate"; title="JSON"; type="application/json", <https://opnsense.org/>; rel=shortlink
Strict-Transport-Security: max-age=15552000; includeSubDomains; preload
Content-Type: text/html; charset=UTF-8
icdfa@icdfa-ns-lab-client-v1:~$
```

Brief Explanation: This means the web request successfully reached the server and received a successful HTTP response.

PART E: OBSERVING PACKETS IN WIRESHARK

5.0 Wireshark Analysis



Brief Explanation: The arp filter displays only arp packet Basically the ubuntu is asking for who has the ip 10.10.10.1 and the firewall LAN interface respond that he is the one this demonstrate the function of ARP which it looks for mac address corresponding to an ip address now both device can communicate at layer 2

5.1 Wireshark Analysis

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.10.10.69	10.10.10.1	ICMP	98	Echo (ping) request id=0x17fd, seq=1/256, ttl=64 (reply in 2)
2	0.000391925	10.10.10.1	10.10.10.69	ICMP	98	Echo (ping) reply id=0x17fd, seq=1/256, ttl=64 (request in 1)
3	1.031533818	10.10.10.69	10.10.10.1	ICMP	98	Echo (ping) request id=0x17fd, seq=2/512, ttl=64 (reply in 4)
4	1.032308422	10.10.10.1	10.10.10.69	ICMP	98	Echo (ping) reply id=0x17fd, seq=2/512, ttl=64 (request in 3)
5	2.033280200	10.10.10.69	10.10.10.1	ICMP	98	Echo (ping) request id=0x17fd, seq=3/768, ttl=64 (reply in 6)
6	2.033912621	10.10.10.1	10.10.10.69	ICMP	98	Echo (ping) reply id=0x17fd, seq=3/768, ttl=64 (request in 5)
7	3.035127749	10.10.10.69	10.10.10.1	ICMP	98	Echo (ping) request id=0x17fd, seq=4/1024, ttl=64 (reply in 8)
8	3.035973394	10.10.10.1	10.10.10.69	ICMP	98	Echo (ping) reply id=0x17fd, seq=4/1024, ttl=64 (request in ...)

Frame 8: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on Ethernet II, Src: VMware_82:fd:7e (00:0c:29:82:fd:7e), Dst: VMware_84:9d:53 (00:0c:29:84:9d:53)
 Destination: VMware_84:9d:53 (00:0c:29:84:9d:53)
 Source: VMware_82:fd:7e (00:0c:29:82:fd:7e)
 Type: IPv4 (0x0800)
 Internet Protocol Version 4, Src: 10.10.10.1, Dst: 10.10.10.69
 Internet Control Message Protocol

Brief Explanation: The icmp filter displays only icmp traffic the ICMP is a protocol use to send an echo request which as we can see the ubuntu send and echo request to the a destination 10.10.10.1 which is the LAN interface of the firewall that connect to the client now the interface send and echo reply to destination address 10.10.10.69 which is the ubuntu

5.1 Wireshark Analysis

No.	Time	Source	Destination	Protocol	Length	Info
11	6.593788634	10.10.10.69	10.10.10.1	DNS	83	Standard query 0x2416 AAAA opnsense.org OPT
12	7.002608320	10.10.10.1	10.10.10.69	DNS	111	Standard query response 0x2416 AAAA opnsense.org AAAA 2001:1af8...

Frame 11: 83 bytes on wire (664 bits), 83 bytes captured (664 bits) on Ethernet II, Src: VMware_84:9d:53 (00:0c:29:84:9d:53), Dst: VMware_82:fd:7e (00:0c:29:82:fd:7e)
 Internet Protocol Version 4, Src: 10.10.10.69, Dst: 10.10.10.1
 User Datagram Protocol, Src Port: 48142, Dst Port: 53
 Domain Name System (query)

Brief Explanation: The dns filter displays only DNS traffic. Now ubuntu sent a DNS AAAA query to 10.10.10.1 requesting for it IPv6 address which is associated with opnsense.org, the DNS resolver returned a response containing the IPv6 address for the domain

WHY IC DFA-NSLAB-CLIENT-V1 USES 10.10.10.1 AS ITS DEFAULT GATEWAY?

ANS: Is because from the lab guid the client has bee configure with

OPNsense LAN IP:

10.10.10.1/24

DHCP range:

10.10.10.100 - 10.10.10.200

Therefore because 10.10.10.1 is the address of the firewall interface connected to the client LAN, When the client needs to communicate with a destination outside its local 10.10.10.0/24 network, it will send the traffic to the firewall at 10.10.10.1, which the traffic will then be routed toward the appropriate network.

QUESTION AND ANSWER

1. What is the difference between the OPNsense WAN and LAN interfaces?

Ans: the LAN interface is used to connect the private network within a limited area which they can communicate through a layer 2 **while** the WAN interface is mainly use to forward the private network data to the internet is the interface that can communicate to the internet directedly

2. Why must both internal adapters use the same VirtualBox network name?

Ans: because the name identifies the **same virtual Layer-2 network** and that is the isolated adapter that connect each of them

3. What information does the default route provide to Ubuntu?

Ans: it provides the ubuntu with information that whenever it received a packet that is does not know it destination it should forward it to the interface

4. Which packet exchange allows Ubuntu to learn the firewall MAC address?

Ans: The Address Resolution Protocol ARP packet exchange look for Mac address corresponding to a given ip address

5. Why does a successful ping to 1.1.1.1 not automatically prove that DNS is working?

Ans: the successfully ping of internet (1.1.1.1) does not prove DNS is working because the DNS is main for mapping a domain name to an ip address and in this case we test the connectivity using ip address not a domain name